Roll No. | | | | | | | | | | | Total No. of Pages : 02

Total No. of Questions : 13

B.Pharmacy (Sem.–1)

PHARMACEUTICAL ANALYSIS-I

Subject Code : BP-102T

M.Code : 74645

Date of Examination : 14-06-2024

Time : 3 Hrs. Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.

SECTION-A

1. **Explain briefly :**

   a) An aqueous solution of *NaCl is neutral but that of* CH_3COONa is alkaline. Why?

   b) Starch cannot be *used as indicator in* KIO_3 titrations. Why?

   c) What type of errors is associated with precision of an experiment? How can these be minimised?

   d) Define solubility product.

   e) Why pH control is important in complexometric titrations?

   f) What is the precise role of mercuric acetate in non-aqueous estimation of ephedrine HCl?

   g) What is the difference between ignition and drying in gravimetric analysis?

   h) Name the reversible indicators used in potassium bromate titrations.

   i) Give balance chemical equations for reaction between $KMnO_4$ and Oxalic acid.

   j) What do you understand by diffusion current?

SECTION-B

2. Plot the neutralization curve by calculating pH at 0, 50, 90, 99, 99.9,100, 100.1,101 and 110 % neutralization reaction between 0.1 M CH_3COOH and 0.1 MNaOH. (pKa of $CH_3COOH = 4.75$) (10)

3. What is the principle of complexometric titrations? Describe its different types with examples. Give the chemical equations involved, method and general calculation for estimation of calcium gluconate. (2+4+4)

4. Describe the concept of oxidation and reduction in pharmaceutical analysis with examples. How do you calculate equivalent weight of an oxidant and a reductant? Write a detailed account on iodimetric and iodometric titrations with emphasis on their principle, applications and limitations. (2+2+6)

SECTION-C

5. Write a detailed account on conductometric titrations.

6. Give the principle, balanced chemical equations and general calculations for the determination of barium as barium sulfate.

7. Describe in detail the Mohr's method.

8. Define the terms primary and secondary standards. Enumerate their characteristic features. Give examples in neutralization, non-aqueous, redox and precipitation titrations.

9. What is the principle of polarographic method of analysis? Describe the construction and working of rotating platinum electrode.

10. Define the terms Colour change interval of an indicator. How is it determined?

11. Describe principle and different methods of end point detection in potentiometric titrations.

12. Give principle, chemical equations, method and general calculations for estimation of sodium benzoate by non-aqueous titrations.

13. How will you prepare a 100 ml solution 0.1 M solution of cerric ammonium sulfate? Describe its methods of standardization emphasizing chemical equations and calculations.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.

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